USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 20 LAB - Array Factor
Lab Sheet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

This sheet is the turn-in document for the L20 Lab. Fill the calculated columns before you sweep and the measured columns at the bench, so the prediction is on paper before the trace is on the screen. Compute every prediction at your own HB100's Signal Freq: use the one your kit reported in the L18 Lab, and recompute if **Find HB100** reports a frequency more than 50 MHz from it at the bench. The lab is individual: one sheet per cadet, from your own kit.

Learning Objective 3.2: I can derive the array factor for an arbitrary linear array and apply pattern multiplication.

The array is the PHASER's: $N = 8$ elements, E1 to E8, on a $d = 14$ mm pitch. Use $c = 3.00 \times 10^8$ m/s.

1. **Your kit's settings.** Record the Signal Freq that **Find HB100** reports and the Rx Gain (dB) that puts the FFT peak at -10 ± 3 dBFS. Leave the gain there for all three sweeps. Then find the wavelength, the electrical spacing, and kd at your Signal Freq.

Signal Freq = _____ Rx Gain = _____

λ = _____ d/λ = _____ kd = _____

2. **The measurement table.** One row per aperture. Fill the calculated columns at your Signal Freq. Report each trace's peak level and its step from the eight-element peak, and report each sidelobe level relative to that trace's own peak.

Active	Peak (dBFS)	Step from 8 (dB)	HPBW meas	HPBW calc	FNBW meas	FNBW calc	First SLL (dBc)
8 (E1–E8)		reference					
4 (E3–E6)							
2 (E4, E5)						180°	none

3. **Written answer 1.** Explain why the first-null beamwidth of the two-element array is quoted as 180° rather than measured. Name the condition on $\lambda/2d$ that decides whether a null exists in visible space, and evaluate it for $d = 14$ mm at your Signal Freq.

4. **Written answer 2.** At boresight the element signals add in voltage, which predicts a fixed peak step every time you halve the number of active elements. State the predicted step at four and at two elements, and compare your two measured steps with it. Leakage through the elements set to 0 and receiver compression were tested on the dry-run kit and ruled out as the main cause of any shortfall. Propose one other possible cause, and describe a measurement on your kit that would test it.

Documentation: _____